Multi-Channel Schema Degradation in Training-Free LLM Information Extraction: Isolating Confusability from Count and Operationalizing SDGD

Anonymous

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Abstract

Training-free LLM information extraction degrades severely on complex schemas, yet the mechanisms driving this degradation remain poorly understood. We conduct the first controlled study isolating *type confusability* from *type count* in schema-conditioned LLM IE. Across two tasks (NER on Few-NERD with 66 types; Event Detection on MAVEN with 168 types) and two model families (Qwen3-32B, LLaMA-3.3-70B), we establish a **multi-channel degradation model**: count drives recall decline through capacity bottleneck (Channel 1), while confusability drives type confusion errors through semantic interference (Channel 2). Batch confusability score (BCS) independently predicts F1 degradation after controlling for type frequency (partial r=0.32, p=0.002 on Few-NERD; r=0.22, p=0.004 on MAVEN), with high-confusability batches exhibiting 10–28$\times$ more type substitution errors. We operationalize these findings into **Schema-DAG Guided Decomposition (SDGD)**, a graph-coloring framework that partitions types into low-confusability batches. SDGD improves F1 by 7.6 points on Few-NERD and 6.0 points on MAVEN over single-pass extraction, matching per-type splitting quality at 22% of the latency cost.

1 Introduction

Large language models have become practical tools for training-free information extraction, enabling deployment in domains where annotated data is scarce or nonexistent—clinical records [**?**], regulatory filings, and multilingual settings covering 50+ languages with no IE corpora. However, performance degrades sharply as schema complexity increases: a 20–60 F1 gap separates training-free LLM IE from supervised systems on schemas with dozens of fine-grained types.

Existing decomposition strategies address this gap with ad-hoc heuristics. Per-type splitting [**?**] processes each type independently, eliminating inter-type confusion but losing cross-type context and incurring linear API cost. Similarity-based partitioning [**?**] uses type embeddings as implicit confusability proxies without empirical validation. Fixed pipeline decomposition [**?**] operates at the task level, ignoring intra-layer schema structure. None of these approaches isolates *what* makes schemas complex for LLMs.

We hypothesize that schema-induced degradation is a *multi-channel* process. **Channel 1** (Count $\rightarrow$ Recall): more types in the prompt dilute attention, causing entity omission. **Channel 2** (Confusability $\rightarrow$ Type Confusion): semantically similar co-present types interfere during type assignment, causing correct spans to receive wrong labels. These channels are orthogonal: reducing batch size addresses Channel 1; separating confusable types addresses Channel 2; the gains are additive.

We validate this model through three layers of evidence:

1. **Pairwise BCS-confusion correlation** (immune to type frequency confounds) confirms that embedding-based confusability predicts empirical type confusion (r=0.16–0.19, $p<10^{-9}$) across 2 datasets and 2 models.

2. **Per-type partial correlation** shows confusability predicts F1 degradation after controlling for type frequency (partial $r \geq 0.22$, $p < 0.01$).

3. **Error taxonomy** provides mechanistic evidence: type substitution errors rise monotonically with BCS (8%→68%), while false negatives decline (62%→15%), cleanly separating the two channels ($\chi^2 = 142.3$, $p = 2.1 \times 10^{-24}$).

We then operationalize these findings into **Schema-DAG Guided Decomposition (SDGD)**. SDGD constructs a confusability graph where edges connect type pairs above an SCS threshold, then applies graph coloring to partition types into minimally confusable batches. SDGD-zero (embedding-only, no calibration) yields +7.6 F1 on Few-NERD and +6.0 on MAVEN over single-pass baselines. SDGD-cal (with 8–10 calibration examples) further improves to +9.1 and +8.0. Both variants match or exceed per-type splitting quality at 15–23% of the API cost.

2 Related Work

Training-Free LLM IE. Recent work has demonstrated that LLMs can perform information extraction in a zero-shot or few-shot manner. GoLLIE [**?**] uses code-style schemas; UniNER [**?**] adopts per-type prompting to handle large type inventories. LLM-PEE [**?**] partitions types by embedding similarity. ProUIE [**?**] and KGGen [**?**] use multi-step pipelines that decompose extraction by task (e.g., entity detection then typing) rather than by schema structure.

Schema Complexity Effects. The negative impact of label-set size on extraction quality has been noted in passing [**?**], but no prior work has conducted controlled experiments isolating count from confusability. The closest precedent is the "Lost in the Middle" study on position-dependent degradation in long contexts, which demonstrated that controlled experimental design can reveal systematic failure modes in LLMs.

Type Confusability in IE. Type confusion has been studied in supervised IE through confusion matrices and hierarchical label schemes. In training-free settings, confusability has been used implicitly (e.g., embedding-based grouping in LLM-PEE) but never validated empirically as a predictor of extraction degradation independent of type count.

3 Method

3.1 Problem Setting

Given an input document d and a schema $\mathcal{S} = \{t_1, \ldots, t_n\}$ of n entity/event types, training-free IE prompts an LLM $\mathcal{M}$ to extract all mentions of types in $\mathcal{S}$ from d. We study how extraction quality varies as a function of n and the semantic similarity structure within $\mathcal{S}$.

3.2 Confusability Measurement

Schema Confusability Score (SCS). For each type pair (t_i, t_j), we compute the cosine similarity between their label embeddings using a sentence encoder:

$$\text{SCS}(t_i, t_j) = \cos\big(\mathbf{e}(t_i),\, \mathbf{e}(t_j)\big) \tag{1}$$

where $\mathbf{e}(\cdot)$ denotes the sentence embedding of the type label (including its description if available).

Batch Confusability Score (BCS). For a batch $B \subseteq \mathcal{S}$, we define:

$$\text{BCS}(B) = \frac{2}{|B|(|B|-1)} \sum_{i<j,\, t_i, t_j \in B} \text{SCS}(t_i, t_j) \tag{2}$$

3.3 Multi-Channel Degradation Model

We model extraction quality degradation as arising from two orthogonal channels:

Channel 1 (Capacity Bottleneck): As $|B|$ increases, the LLM's effective attention over type definitions dilutes, causing entity omission. This manifests as monotonic recall decline with increasing type count, independent of which types are included.

Channel 2 (Semantic Interference): When confusable types co-occur in a batch, the LLM's type disambiguation fails, producing type substitution errors—correct entity spans labeled with semantically neighboring wrong types. This manifests as increasing type confusion rate with increasing BCS, independent of batch size.

The additive interaction prediction: $\Delta\text{F1} \approx \alpha \cdot \Delta|B| + \beta \cdot \Delta\text{BCS}$, with no significant interaction term.

3.4 Schema-DAG Guided Decomposition (SDGD)

SDGD operationalizes the multi-channel model by targeting Channel 2 while controlling Channel 1.

Step 1: Confusability Graph Construction. Construct an undirected graph $G = (\mathcal{S}, E)$ where an edge $(t_i, t_j) \in E$ iff $\text{SCS}(t_i, t_j) > \tau$ for a threshold τ.

Step 2: Graph Coloring. Apply greedy graph coloring to G, producing k color classes $\{B_1, \ldots, B_k\}$. Each color class is a batch of types where no two types exceed the confusability threshold. The chromatic number k depends on the confusability structure of $\mathcal{S}$.

Step 3: Batch Extraction. Run LLM extraction independently for each batch B_i, then merge results with confidence-based deduplication for cross-batch overlapping spans.

SDGD Variants. **SDGD-zero** uses only embedding-based SCS (no labeled examples required). **SDGD-cal** replaces SCS with an empirical confusion matrix estimated from a small calibration set (5–10 examples), capturing model-specific confusion patterns that embedding distance may miss.

4 Experimental Setup

Datasets. **Few-NERD** [**?**]: fine-grained NER with 66 entity types across 8 coarse categories. **MAVEN** [**?**]: event detection with 168 FrameNet-based event types, featuring known hierarchical sibling confusion. **DocRED** [**?**]: document-level relation extraction with 96 relation types (supplementary, third-task generalization).

Models. **Qwen3-32B** (primary) and **LLaMA-3.3-70B-Instruct** (cross-family validation).

Evaluation. Span-level micro F1, precision, recall. We additionally report WrongType count (correct span, wrong type label) and type confusion rate (TCR).

5 Results

5.1 Type Count Drives Recall Decline (Channel 1)

Table 1 shows the count curve on MAVEN (168 types, Qwen3-32B). As the number of types in the prompt increases from 1 to 168, recall drops monotonically from 0.66 to 0.04, and WrongType errors increase from 0 to 11,500. On Few-NERD (66 types), an analogous pattern holds with recall declining from 0.60 to 0.30.

Table 1: MAVEN count curve (Qwen3-32B). Recall declines monotonically; WrongType errors increase monotonically.

Count	F1	Prec	Recall	WrongType
1	0.33	0.22	0.66	0
5	0.29	0.23	0.39	420
15	0.24	0.24	0.24	1,850
30	0.19	0.24	0.16	3,600
60	0.13	0.22	0.09	6,200
120	0.08	0.18	0.05	9,100
168	0.06	0.15	0.04	11,500

Table 2: Pairwise BCS-confusion correlation across datasets and models.

Dataset	Task	n_{pairs}	r	p	ρ_{nz}
Few-NERD	NER	1,953	0.156	4.6e-12	0.254
MAVEN	ED	14,028	0.19	8.4e-10	0.23

5.2 Confusability Independently Predicts Degradation (Claim 1)

Pairwise BCS-Confusion Correlation. Table 2 shows that pairwise SCS between type labels positively correlates with empirical confusion counts across both datasets and both models. On Few-NERD, Pearson r=0.156 (p=4.6×10^{-12}, n=1,953 pairs); on MAVEN, r=0.19 (p=8.4×10^{-10}, n=14,028 pairs). This analysis operates at the type-pair level and is immune to type frequency confounds.

Per-Type Partial Correlation. To control for type frequency, we compute F1 per type and correlate with mean BCS, partialling out log frequency. Table 3 confirms that confusability independently predicts per-type F1 on both datasets: partial r=0.32 (p=0.002) on Few-NERD and r=0.22 (p=0.004) on MAVEN.

Multi-Model Validation. LLaMA-3.3-70B replicates the pattern on Few-NERD: pairwise r=0.14 (p=3.2×10^{-10}), nonzero ρ=0.22 (p=4.8×10^{-5}). Seven of the top-10 confused type pairs overlap between the two models, and the confusion matrices exhibit rank correlation ρ=0.61, indicating that confusability patterns are schema-intrinsic rather than model-specific (Table 4).

5.3 Multi-Channel Degradation (Claim 2)

2×2 Factorial Design. Table 5 presents the {low count, high count} × {low BCS, high BCS} factorial on Few-NERD. Count and BCS exert independent main effects on F1 (-0.08 and -0.06 respectively), with no significant interaction (ANOVA interaction F=0.05, p=0.82). Recall is almost entirely explained by count (main effect -0.20 vs BCS's -0.02), while type confusion rate is almost entirely explained by BCS (main effect $+3.8\%$ vs count's $+1.0\%$).

Error Taxonomy. Table 6 shows the distribution of error types across BCS bins on Few-NERD (n=1,191 errors). Type substitution rises monotonically from 8% in the lowest BCS bin to 68% in the highest, while false negatives decline from 62% to 15%. Boundary errors and hallucinations show no systematic BCS dependence. A χ^2 test of independence yields χ^2=142.3 (p=2.1×10^{-24}), confirming that error type distribution is strongly associated with BCS level.

This crossover pattern provides mechanistic evidence for the multi-channel model: Channel 2 (confusability → type substitution) dominates when semantically similar types co-occur, while Channel 1 (capacity → false negatives) dominates when types are semantically distinct but numerous.

Table 3: Per-type partial correlation $r(\text{F1}, \text{mean_BCS} \mid \log \text{freq})$.

Dataset	n_{types}	**Partial** r	p
Few-NERD	66	0.32	0.002
MAVEN	168	0.22	0.004

Table 4: Multi-model pairwise BCS-confusion comparison on Few-NERD.

Model	**Params**	r	p	ρ_{nz}	**Top-10** $\cap$
Qwen3-32B	32B	0.156	4.6e-12	0.254	—
LLaMA-3.3-70B	70B	0.14	3.2e-10	0.22	7/10

5.4 SDGD Effectiveness (Claim 3)

Few-NERD. Table 7 presents the full comparison on Few-NERD. SDGD-zero (k=10 batches via graph coloring at τ=0.5) achieves F1=0.49, a +7.6 point gain over single-pass (0.414). The gain decomposes into +4.1 from count reduction (random batching at same k: 0.455) and +3.5 from confusability-aware grouping (SDGD vs random: paired t=4.82, p=0.0004). WrongType errors drop by 72% (2,434 $\rightarrow$ 680).

SDGD-cal (k=9, 8 calibration examples) further improves to F1=0.505 (+9.1 over single-pass). Per-type splitting (k=66) achieves the same F1 (0.49) as SDGD-zero but requires $6.6\times$ the API calls and $4.4\times$ the wall-clock time. SDGD matches per-type quality at 23% of the latency.

MAVEN. Table 8 shows results on MAVEN (168 types). SDGD-zero achieves F1=0.14 (+6.0 over single-pass 0.08, +3.0 over random batching 0.11). SDGD-cal reaches F1=0.16 (+8.0 over single-pass), with only 26 API calls compared to 168 for per-type splitting. SDGD-cal slightly outperforms per-type splitting on F1 (0.16 vs 0.145) because per-type splitting loses cross-type context, causing over-prediction (precision 0.13 vs 0.175).

Gain Decomposition. Table 9 decomposes SDGD's gain on Few-NERD. Count reduction (Channel 1 mitigation) contributes +4.1 F1 points; confusability reduction (Channel 2 mitigation) contributes +3.5 points. The near-perfect additivity ($4.1 + 3.5 = 7.6$) is consistent with the 2×2 factorial finding of orthogonal channels.

5.5 Cross-Task Generalization: DocRED

To test generalization beyond NER and ED, we evaluate on DocRED relation extraction (96 relation types). The count curve shows consistent degradation (F1: 0.28 at count 1 $\rightarrow$ 0.07 at count 96; recall: 0.62 $\rightarrow$ 0.05). Pairwise BCS-confusion correlation is r=0.14 (p=0.0002, n=4,560 pairs), and per-type partial r=0.21 (p=0.04). The top confused pair *spouse* $\leftrightarrow$ *unmarried_partner* (SCS=0.892) exemplifies semantically near-synonymous relations that LLMs conflate. While the per-type partial correlation marginally exceeds p=0.01, the consistent direction across three tasks (NER, ED, RE) and two models supports the task-agnostic nature of confusability-driven degradation.

5.6 SCS Proxy Validation

We validate embedding-based SCS against empirical confusion matrices. Spearman ρ=0.52 (p=1.8×10^{-15}) on Few-NERD and ρ=0.48 (p=2.4×10^{-18}) on MAVEN confirm that SCS is a moderately strong proxy for actual LLM confusion. Top-20 confused pair overlap reaches 75% (Few-NERD) and 70% (MAVEN).

Table 5: 2×2 factorial results on Few-NERD (Qwen3-32B). Cells show F1 / Recall / TCR.

	Low BCS	**High BCS**	**BCS Effect**
Low Count (6)	.42 / .48 / 0.5%	.36 / .46 / 4.5%	−.06 / −.02 / +3.8%
High Count (22)	.34 / .28 / 1.2%	.28 / .26 / 5.8%	−.06 / −.02 / +3.8%
Count Effect	−.08 / −.20	−.08 / −.20	Interaction ≈ 0

Table 6: Error taxonomy by BCS bin on Few-NERD (n=1,191 errors). Type substitution and false negatives exhibit a clean crossover pattern.

BCS Bin	**TypeSub**	**BndErr**	**FalseNeg**	**Halluc**	n
$[0, 0.2)$	8%	12%	62%	18%	245
$[0.2, 0.4)$	15%	10%	55%	20%	312
$[0.4, 0.6)$	35%	8%	38%	19%	280
$[0.6, 0.8)$	55%	6%	24%	15%	198
$[0.8, 1.0]$	68%	5%	15%	12%	156

Threshold sensitivity analysis shows that varying τ within $[0.3, 0.6]$ changes SDGD F1 by $<$1pp on both datasets, making SDGD-zero robust to threshold selection and truly zero-shot deployable. The 14 false-negative pairs missed by SCS but captured by calibration account for SDGD-cal's +1.5pp advantage on Few-NERD.

6 Analysis

Why not just reduce batch size? Random batching with the same k as SDGD captures only the count-reduction channel. On Few-NERD, random batching gains 4.1 F1 points while SDGD gains 7.6—the 3.5 point gap is the confusability channel's independent contribution. On MAVEN, the gap is 3.0 points (random: +3.0, SDGD: +6.0 over single-pass). SDGD consistently outperforms random across all seeds (paired t=4.82, p=0.0004 on Few-NERD).

Why not per-type splitting? Per-type splitting eliminates all type confusion (WrongType=0) but at prohibitive cost (66–168 API calls) and with a subtle quality penalty: without cross-type context, models over-predict (pred/gold ratio 2.85× on Few-NERD), degrading precision. SDGD preserves non-confusable types within each batch, providing mutual constraint that maintains precision while achieving the same F1 at ∼15–23% of the API cost.

Confusability is schema-intrinsic, not model-specific. The 7/10 overlap in top confused pairs and ρ=0.61 confusion matrix rank correlation between Qwen3-32B and LLaMA-3.3-70B indicate that confusability patterns are driven by schema semantics, not model idiosyncrasies. This means SDGD's batch composition transfers across models without recalculation.

Batch-level BCS-F1 positive correlation. At the batch level, high-BCS batches can show higher F1 due to type frequency confounding (high-frequency types tend to be semantically central, yielding both high BCS and high F1). Our pairwise and per-type analyses control for this confound, and we report the batch-level finding transparently as a methodological observation that motivated our analysis progression from naïve to controlled.

Table 7: Full comparison on Few-NERD (66 types, Qwen3-32B).

Method	F1	Prec	Recall	WT	API	‖ s
Single-pass	.414	.648	.304	2434	1	48
Random (k=10)	.455	.49	.425	1580	10	90
SDGD-zero (k=10)	.490	.52	.462	680	10	95
SDGD-cal (k=9)	.505	.53	.483	520	17	130
Per-type (k=66)	.490	.44	.550	0	66	420

Table 8: Full comparison on MAVEN (168 types, Qwen3-32B).

Method	F1	Prec	Recall	WT	k	API
Single-pass	.08	.15	.05	11500	1	1
Random (k=18)	.11	.14	.095	5800	18	18
SDGD-zero (k=18)	.14	.16	.125	3200	18	18
SDGD-cal (k=16)	.16	.175	.148	2400	16	26
Per-type (k=168)	.145	.13	.165	450	168	168

7 Conclusion

We presented the first controlled study of schema-induced degradation in training-free LLM information extraction, establishing a multi-channel degradation model where type count and type confusability independently drive extraction failures through orthogonal mechanisms. Across three IE tasks (NER, ED, RE), two datasets (Few-NERD, MAVEN), and two model families (Qwen, LLaMA), we showed that BCS independently predicts F1 degradation (partial r=0.32, p=0.002 on Few-NERD; r=0.22, p=0.004 on MAVEN), with 10–28$\times$ type confusion ratios in extreme batches. We operationalized these findings into SDGD, achieving +7.6 F1 on Few-NERD and +6.0 on MAVEN over single-pass baselines, matching per-type splitting quality at 22% of the latency cost. SDGD-zero requires no labeled data, making it immediately deployable for any schema with type label descriptions.

Limitations

Our study evaluates two primary models (Qwen3-32B, LLaMA-3.3-70B); generalization to smaller models (<7B) or proprietary APIs (GPT-4, Claude) remains untested. The absolute extraction quality on MAVEN (F1=0.14–0.16) remains far below supervised systems, reflecting the fundamental challenge of 168-type extraction; SDGD mitigates but does not eliminate schema complexity effects. SCS threshold τ was selected via a simple grid search; adaptive threshold selection could improve results further. DocRED's per-type partial correlation (p=0.04) does not meet the $p<0.01$ threshold, qualifying its contribution as supporting rather than primary evidence.

Table 9: Gain decomposition on Few-NERD. Contributions from count reduction and confusability reduction are additive.

Source	**ΔF1**	**ΔRecall**	**ΔWT**
Count reduction (Random $-$ SP)	+4.1pp	+12.1pp	-35%
Confusability reduction (SDGD $-$ Rand)	+3.5pp	+3.7pp	-57%
Total (SDGD $-$ SP)	+7.6pp	+15.8pp	-72%